TECHNICAL DOCUMENTATION

Tilt Sensor Activated Alarm System

with Relay, Buzzer and LED

Project Level: 2.6

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1. Project Overview

This project implements a tilt-activated alarm system using a SW-520 tilt sensor as a digital trigger. When the device is tilted, the sensor closes, activating a transistor-driven relay that simultaneously turns on a buzzer and a red LED. When the device returns to its flat position, the alarm is silenced and the LED turns off.

The circuit demonstrates fundamental concepts in digital switching, transistor saturation, relay-based load control, and inductive protection — all at a level appropriate for middle school electronics students.

2. Component Table

Component	Value / Model	Qty	Role in Circuit
SW-520 Tilt Sensor	SW-520	1	Acts as a digital switch; closes when tilted, opens when flat
NPN Transistor	2N2222	1	Amplifies base signal to drive relay coil; acts as electronic switch
Base Resistor	1 kΩ	1	Limits current flowing into transistor base, prevents overdrive
Pull-Down Resistor	10 kΩ	1	Keeps base voltage at 0 V (GND) when tilt sensor is open
Relay	LU-5-R (9 V)	1	Electrically isolates low-power control circuit from load; connects buzzer and LED
Flyback Diode	1N4148	1	Absorbs back-EMF spike from relay coil when transistor turns off
Red LED	5 mm Red	1	Visual indicator; turns on when tilt is detected
LED Resistor	470 Ω	1	Limits LED current to a safe level (~17.6 mA)
Buzzer	Active 5 V / 9 V	1	Audio alarm; activates when relay closes
9 V Battery	9 V DC	1	Powers the entire circuit including relay coil, buzzer, and LED

3. Detailed Component Descriptions

3.1 SW-520 Tilt Sensor

The SW-520 is a simple electromechanical tilt sensor that contains a small metallic ball inside a conductive cylinder. When the sensor is in its upright (flat) position, the ball rests away from the contacts and the circuit is open (no current flows). When the sensor is tilted beyond a threshold angle (approximately 45°), the ball rolls down and bridges the two internal contacts, closing the circuit.

- Operating voltage: 3 V – 12 V DC
- Contact resistance (closed): < 100 Ω
- No external power required — purely mechanical switch
- Typical operating life: > 100,000 cycles

3.2 NPN Transistor (2N2222)

The 2N2222 is a general-purpose NPN bipolar junction transistor (BJT). In this circuit it operates as a switch between two states: cutoff (OFF) and saturation (ON). When a sufficient base current is supplied, the transistor saturates and allows the collector-emitter path to conduct, energizing the relay coil.

- Package: TO-92 (3 pins: Emitter, Base, Collector)
- Maximum collector current (IC max): 600 mA
- Maximum collector-emitter voltage (VCE max): 40 V
- DC current gain (hFE): 100 – 300 (typical 200)
- VBE (saturation): ~0.7 V

3.3 Relay (LU-5-R, 9 V Coil)

The LU-5-R is an electromechanical relay with a 9 V DC coil. When sufficient current flows through the coil, it generates a magnetic field that physically moves an armature, switching the contact set. The relay provides galvanic isolation between the low-power control side (transistor, tilt sensor) and the load side (buzzer, LED).

- Coil voltage: 9 V DC
- Coil resistance: ~200 Ω (measured typical)
- Contact types: NO (Normally Open), NC (Normally Closed), COM (Common)
- Contact rating: 3 A / 125 V AC ; 3 A / 24 V DC

In this project the NO (Normally Open) and COM contacts are used. When the relay activates, COM connects to NO, completing the load circuit.

3.4 Flyback Diode (1N4148)

The relay coil is an inductor. When the transistor suddenly cuts off current to the coil, the collapsing magnetic field induces a large reverse-polarity voltage spike (back-EMF). Without protection this spike can exceed the transistor's VCE max (40 V) and permanently damage it.

The 1N4148 fast-switching diode is placed in parallel with the relay coil, oriented in reverse bias during normal operation. When the spike occurs, the diode becomes forward biased and provides a low-resistance path for the energy to dissipate safely through the coil resistance.

- Forward voltage: 0.7 V
- Reverse breakdown: 100 V
- Peak current: 450 mA (well above any spike from this relay)
- Switching speed: < 4 ns — fast enough to catch inductive spikes

3.5 Red LED

A standard 5 mm through-hole red LED is used as the visual alarm indicator. The LED is a semiconductor diode that emits red light (wavelength ~620–630 nm) when forward biased with adequate current.

- Forward voltage (VF): ~2.0 V
- Recommended forward current (IF): 10 – 20 mA
- Absolute maximum current: 30 mA

3.6 Buzzer (Active)

An active buzzer contains an internal oscillator circuit. It produces a continuous audible tone simply by applying the rated DC voltage across its terminals — no external frequency signal is needed. This makes it ideal for alarm applications.

- Operating voltage: 5 V or 9 V (depending on module)
- Operating current: typically 30 – 50 mA
- Sound pressure level: ≥85 dB at 10 cm

4. Circuit Calculations

4.1 Relay Coil Current

The relay coil draws current from the 9 V supply through the transistor collector-emitter path.

```
V_supply = 9 V
R_coil   = 200 Ω  (LU-5-R typical)
VCE(sat) = 0.2 V  (transistor in saturation)

I_coil = (V_supply - VCE(sat)) / R_coil
I_coil = (9 - 0.2) / 200
I_coil = 8.8 / 200
I_coil ≈ 44 mA
```

This collector current (IC) becomes the reference for base resistor sizing.

4.2 Transistor Saturation Check

For the transistor to reliably saturate, the base current must be sufficient. We use a saturation factor of 10 to guarantee hard saturation over component tolerances.

```
hFE (minimum) = 100  (2N2222 worst-case)
IC = 44 mA

IB_min = IC / hFE = 44 / 100 = 0.44 mA
IB_required = IB_min × 10 (saturation factor)
IB_required = 0.44 × 10 = 4.4 mA
```

We will verify that the chosen 1 kΩ base resistor provides at least this much base current.

4.3 Base Resistor Calculation (1 kΩ)

When the tilt sensor closes, the sensor connects V_supply (9 V) to the base resistor network. The base resistor limits the current entering the transistor base.

```
V_supply = 9 V
VBE      = 0.7 V  (base-emitter junction drop)
R_base   = 1 kΩ = 1000 Ω

IB = (V_supply - VBE) / R_base
IB = (9 - 0.7) / 1000
IB = 8.3 / 1000
IB = 8.3 mA
```

Since IB (8.3 mA) >> IB_required (4.4 mA), the transistor is guaranteed to saturate fully. The 1 kΩ choice is correct.

4.4 Pull-Down Resistor Explanation (10 kΩ)

When the tilt sensor is open (flat position), the base pin would be left floating without the pull-down resistor. A floating base can pick up electromagnetic noise and cause spurious transistor triggering.

The 10 kΩ resistor connects the base to GND, ensuring the base voltage is firmly held at 0 V when the sensor is open. The high value (10 kΩ) is chosen because:

- It draws minimal current from the supply (only 0.9 mA if V_supply were applied through it)
- It is much larger than R_base (1 kΩ), so it does not significantly affect IB when the sensor closes
- It reliably overcomes any small leakage currents or induced noise

4.5 LED Current Limiting Resistor (470 Ω)

The LED and its series resistor are connected across the relay NO-COM contacts, powered by the 9 V supply through the relay.

```
V_supply = 9 V
VF_LED   = 2.0 V  (red LED forward voltage)
R_LED    = 470 Ω

I_LED = (V_supply - VF_LED) / R_LED
I_LED = (9 - 2.0) / 470
I_LED = 7.0 / 470
I_LED ≈ 14.9 mA
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14.9 mA is within the safe range (10–20 mA) for a standard red LED. The LED will illuminate brightly without risk of damage.

5. Pin-by-Pin Connection Diagram

The following table describes every wire connection in the circuit. Use this as a checklist when building on a breadboard.

#	From	To	Notes
1	9 V Battery (+)	Breadboard (+) rail	Main positive supply
2	9 V Battery (-)	Breadboard (-) rail	Ground / common negative
3	(+) rail	SW-520 pin 1	Supplies voltage to tilt sensor
4	SW-520 pin 2	Junction: R_base (1 kΩ) + R_pulldown (10 kΩ)	Sensor output node
5	R_base (1 kΩ) other end	2N2222 Base (pin 2)	Base drive current path
6	R_pulldown (10 kΩ) other end	GND (-) rail	Holds base at 0 V when sensor open
7	2N2222 Emitter (pin 1)	GND (-) rail	Emitter always to ground in NPN switch
8	2N2222 Collector (pin 3)	Relay coil pin A	Collector drives coil current
9	Relay coil pin B	(+) rail (9 V)	Other end of coil to positive supply
10	1N4148 Anode	Collector / Relay coil pin A	Diode anode at lower potential
11	1N4148 Cathode	Relay coil pin B / (+) rail	Diode cathode at higher potential
12	Relay COM	(+) rail (9 V)	Common contact powered from supply
13	Relay NO	R_LED (470 Ω) pin 1 + Buzzer (+)	Load power through NO contact
14	R_LED (470 Ω) pin 2	LED Anode (+)	Current-limited LED supply
15	LED Cathode (-)	GND (-) rail	LED return path
16	Buzzer (-)	GND (-) rail	Buzzer return path

6. Working Principle

6.1 When Device is FLAT (Alarm OFF)

The SW-520 tilt sensor is in its upright position. The internal ball is away from the contacts, so the sensor acts as an open circuit. The 10 kΩ pull-down resistor holds the transistor base pin firmly at 0 V (GND). With VBE = 0 V, the transistor is in cutoff mode — no collector current flows. The relay coil is de-energized, the armature is held open by its spring, and the NO contact remains open. Neither the buzzer nor the LED receives power. The circuit is silent.

6.2 When Device is TILTED (Alarm ON)

1. The SW-520 tilts beyond its threshold angle.
2. The internal ball rolls and bridges the two contacts — the sensor closes.
3. Current flows: 9 V → SW-520 → Junction node → 1 kΩ R_base → Base of 2N2222 → Emitter → GND.
4. IB ≈ 8.3 mA is injected into the base, saturating the transistor.
5. With transistor saturated, the collector-emitter path has very low resistance (~0.2 V drop). Current flows: 9 V → Relay coil pin B → Relay coil → Relay coil pin A → Collector → Emitter → GND.
6. IC ≈ 44 mA energizes the relay coil, creating a magnetic field.
7. The magnetic field pulls the armature, connecting COM to NO.
8. Current flows through NO-COM: 9 V → COM → NO → R_LED (470 Ω) → LED → GND (LED turns ON).
9. Simultaneously: 9 V → NO → Buzzer (+) → Buzzer internal oscillator → Buzzer (-) → GND (Buzzer sounds).

6.3 When Device Returns to FLAT (Alarm OFF)

10. Tilt sensor opens again — ball rolls away from contacts.
11. Base voltage drops to 0 V via pull-down resistor.
12. Transistor enters cutoff. Collector current suddenly stops.
13. The relay coil's collapsing magnetic field generates a back-EMF spike.
14. The 1N4148 flyback diode clamps this spike by conducting the induced current in a loop around the coil, protecting the transistor.
15. Relay armature springs back, COM disconnects from NO. Both buzzer and LED turn off.

7. Possible Problems and Solutions

Problem	Likely Cause	Solution
Buzzer/LED always ON even when flat	Pull-down resistor missing or not connected to GND	Verify 10 kΩ from base node to GND rail
Buzzer/LED never turns ON when tilted	Transistor not saturating; wrong resistor values; bad connections	Check IB path continuity; verify 1 kΩ at base; test sensor with multimeter
Transistor gets hot quickly	Not saturating fully; collector-emitter high resistance	Check base current; ensure 2N2222 is correctly oriented (flat face forward)
Relay clicks but buzzer/LED stay off	Load wired to NC instead of NO; or loose wire	Confirm wires go to NO and COM pins, not NC
Inconsistent triggering / chattering	Sensor vibration or marginal threshold angle	Physically secure sensor; add small 100 nF decoupling cap across base-GND for debounce
Transistor burns out after a few uses	Flyback diode missing or reversed	Check 1N4148 orientation: cathode (banded end) to relay coil (+) side
LED too dim	LED resistor too large, or LED connected backwards	Try 220 Ω for brighter light; check LED polarity (longer leg = anode)

8. Safety Notes

8.1 Electrical Safety

- This circuit operates at 9 V DC from a battery — this voltage is safe to handle. However, always disconnect the battery before modifying the circuit to avoid short circuits.
- Never connect the relay to mains (AC household) power without proper insulation, certified enclosure, and adult supervision. The relay contacts are rated for AC loads but the breadboard circuit is not designed for mains voltage.
- Check all connections before applying power. A short circuit between the 9 V rail and GND will drain the battery rapidly and can cause components to overheat.

8.2 Component Handling

- The 2N2222 transistor pins are close together. Use a magnifying glass or multimeter continuity test to verify correct pin insertion (Emitter / Base / Collector from left to right when flat face is toward you).
- LEDs are polarity-sensitive. Inserting backwards will prevent them from lighting but will not damage them at 9 V.
- The 1N4148 diode is polarity-critical. A reversed flyback diode shorts the relay coil and will cause excessive current draw or transistor failure.
- Avoid bending transistor and diode leads sharply at the body — leave at least 2 mm before the first bend.

8.3 Battery Safety

- Use a fresh alkaline 9 V battery. A low battery may cause the relay to chatter (rapidly open and close) because coil voltage drops below the pull-in threshold.
- Remove the battery when the project is not in use to prevent slow drain through the pull-down resistor.
- Do not short the battery terminals together. A 9 V battery can deliver enough current to cause a wire or small component to heat up quickly.

8.4 Heat and Environment

- If any component becomes hot to the touch within a few seconds of power-on, immediately disconnect the battery and recheck your wiring.
- Keep the circuit away from water, conductive surfaces, and metal objects.
- The active buzzer can produce sound levels above 85 dB. Do not hold the buzzer directly against your ear.

9. Key Terms Glossary

Term	Definition
NPN Transistor	Bipolar junction transistor where a small base current controls a larger collector current. Conducts when base voltage > ~0.7 V.
Saturation	State where the transistor is fully ON; VCE is minimal (~0.2 V). Achieved when IB is large enough relative to IC / hFE.
Cutoff	State where the transistor is fully OFF; no collector current flows. Occurs when VBE < 0.7 V.
hFE / Beta	DC current gain of a transistor. Ratio of IC to IB. Higher hFE means less base current needed.
Back-EMF	Reverse voltage spike generated by an inductor (relay coil) when current through it is suddenly interrupted.
Pull-Down Resistor	Resistor connecting a signal node to GND to define a default LOW state when no driver is active.
NO Contact	Normally Open relay contact: open (disconnected) when relay is de-energized, closed when energized.
Forward Voltage (VF)	Voltage drop across a diode or LED when it conducts in the forward direction.
DXA	Document eXtended unit of measurement in Word documents; 1440 DXA = 1 inch.